

PROJECT_CONTEXT.md

NarrativeIQ

Global Event & Narrative Intelligence Platform

Version: 1.0 Project Type: Data Warehousing & Big Data Semester Project Target Outcome: Exhibition-Level Intelligence Platform Development Strategy: Production-Grade Architecture

1. Project Vision

NarrativeIQ is an advanced intelligence platform designed to transform large-scale fragmented digital information into structured narrative intelligence.

The platform collects information from multiple digital sources, processes and enriches the data through ETL pipelines, stores historical information within a dimensional data warehouse, and generates actionable intelligence regarding event evolution, narrative formation, sentiment shifts, topic growth, and influence propagation.

Unlike traditional dashboards that focus on isolated metrics, NarrativeIQ focuses on reconstructing how narratives emerge, evolve, peak, and decline over time.

The system should feel closer to a professional intelligence platform than a student project.

2. Core Philosophy

Most analytics systems answer:

"What happened?"

NarrativeIQ answers:

"How did it evolve?"

Every module should support this philosophy.

Narratives are first-class entities.

The platform must preserve historical context and support longitudinal analysis.

3. High-Level Architecture

Digital Sources ↓ Data Collection Layer ↓ Raw Data Lake ↓ ETL & Enrichment Pipeline ↓ Enterprise Data Warehouse ↓ Narrative Intelligence Engine ↓ Predictive Analytics Engine ↓ Visualization Portal

4. Development Goals

Primary Goals:

- Demonstrate Data Warehousing concepts
- Demonstrate Big Data concepts
- Implement full ETL lifecycle
- Build dimensional warehouse models
- Provide historical intelligence
- Support predictive analytics
- Deliver exhibition-quality visualization

Secondary Goals:

- Modular architecture
 - Scalable design
 - Production-ready code structure
 - Clean API architecture
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5. System Modules

Module 1: Data Collection Layer

Module 2: Data Lake Layer

Module 3: ETL Pipeline

Module 4: Warehouse Layer

Module 5: Narrative Intelligence Engine

Module 6: Entity Intelligence Engine

Module 7: Sentiment Intelligence Engine

Module 8: Predictive Analytics Engine

Module 9: Visualization Portal

Module 10: Administration Layer

6. Data Sources

Initial Data Sources:

- News datasets
- Event datasets
- Social discussion datasets
- Public forum datasets
- Kaggle datasets

Future Sources:

- Live APIs
- RSS Feeds
- Real-time streams

The architecture must support future source expansion.

7. Data Lake Design

Purpose:

Store raw information before transformation.

Tables:

raw_articles

raw_posts

raw_comments

raw_events

raw_import_logs

Requirements:

- Immutable storage
 - Timestamp preservation
 - Source tracking
 - Import auditing
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8. ETL Pipeline

Stage 1: Extraction

Responsibilities:

- Import datasets
 - Validate files
 - Record source metadata
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Stage 2: Cleaning

Responsibilities:

- Remove duplicates
 - Remove malformed records
 - Standardize timestamps
 - Normalize text
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Stage 3: Transformation

Responsibilities:

- Topic assignment
 - Category assignment
 - Event linking
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Stage 4: Enrichment

Responsibilities:

- Entity extraction
- Sentiment analysis

- Keyword extraction
 - Trend scoring
-

Stage 5: Warehouse Loading

Responsibilities:

- Update dimensions
 - Insert fact records
 - Refresh aggregates
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9. Warehouse Architecture

Design Pattern:

Star Schema

Purpose:

Optimized analytical processing.

10. Dimension Tables

Dim_Date

Fields:

- date_key
 - day
 - month
 - quarter
 - year
-

Dim_Source

Fields:

- source_key
- source_name
- source_type

Dim_Event

Fields:

- event_key
- event_name
- category
- region

Dim_Entity

Fields:

- entity_key
- entity_name
- entity_type

Entity Types:

- Person
- Organization
- Product
- Location

Dim_Topic

Fields:

- topic_key
- topic_name
- topic_category

Dim_Sentiment

Fields:

- sentiment_key
- sentiment_label

Values:

- Positive

- Neutral
 - Negative
-

11. Fact Tables

Fact_Content

Measures:

- engagement_score
 - popularity_score
 - reach_score
-

Fact_Sentiment

Measures:

- sentiment_score
 - polarity
-

Fact_Entity_Mentions

Measures:

- mention_count
 - mention_strength
-

Fact_Narrative

Measures:

- narrative_strength
 - narrative_growth
 - influence_score
-

Fact_Trends

Measures:

- trend_score
 - growth_rate
 - momentum
-

12. Materialized Views

MV_TopNarratives

MV_TopEntities

MV_DailySentiment

MV_TopicGrowth

MV_InfluenceRankings

Purpose:

Fast dashboard rendering.

13. Narrative Intelligence Engine

Purpose:

Identify and reconstruct narratives.

Capabilities:

- Narrative discovery
- Narrative clustering
- Narrative evolution tracking
- Narrative lifecycle analysis

Lifecycle States:

- Emerging
- Growing
- Peak
- Declining
- Archived

14. Entity Intelligence Engine

Purpose:

Track influential entities.

Capabilities:

- Entity ranking
- Relationship analysis
- Mention frequency analysis
- Entity trend analysis

Outputs:

- Top entities
 - Entity timelines
 - Relationship graphs
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15. Sentiment Intelligence Engine

Purpose:

Analyze emotional direction.

Capabilities:

- Sentiment scoring
 - Sentiment trends
 - Volatility analysis
 - Sentiment forecasting
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16. Predictive Analytics Engine

Purpose:

Forecast future narrative behavior.

Predictions:

- Topic growth
- Topic decline
- Narrative popularity
- Sentiment direction

Algorithms:

Phase 1:

- Moving averages
- Trend extrapolation

Phase 2:

- Machine learning models
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17. Dashboard Architecture

Dashboard 1:

Executive Intelligence Dashboard

Widgets:

- Active Narratives
 - Trending Topics
 - Sentiment Overview
 - Top Entities
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Dashboard 2:

Narrative Explorer

Features:

- Narrative timeline
 - Growth visualization
 - Lifecycle stage
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Dashboard 3:

Entity Intelligence

Features:

- Entity rankings
 - Relationship graph
 - Mention trends
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Dashboard 4:

Sentiment Intelligence

Features:

- Sentiment distribution
 - Sentiment evolution
 - Sentiment volatility
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Dashboard 5:

Predictive Intelligence

Features:

- Future growth forecasts
 - Emerging topics
 - Risk indicators
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18. API Architecture

Backend Framework:

FastAPI

API Groups:

/auth

/sources

/events

/narratives

/entities

/sentiment

/predictions

/dashboard

/admin

REST-first architecture.

19. Frontend Architecture

Framework:

Next.js

Language:

TypeScript

Styling:

TailwindCSS

Charts:

Recharts

Advanced Visualizations:

D3.js

Pages:

Home

Dashboard

Narratives

Events

Entities

Predictions

Administration

20. Security Model

Roles:

Admin

Analyst

Viewer

Capabilities:

Role-based permissions

Protected APIs

Audit logging

21. Development Roadmap

Phase 1

Project Foundation

- Database
 - Backend setup
 - Frontend setup
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Phase 2

Data Pipeline

- ETL
 - Cleaning
 - Loading
-

Phase 3

Warehouse

- Star schema
 - Fact tables
 - Dimensions
-

Phase 4

Intelligence Engines

- Narrative engine
 - Sentiment engine
 - Entity engine
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Phase 5

Dashboards

- Visualizations
 - Analytics
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Phase 6

Prediction Layer

- Forecasting
 - Trend detection
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Phase 7

Optimization

- Performance tuning
 - UI refinement
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22. Exhibition Strategy

The project must be demonstrated as an intelligence platform.

Avoid describing it as:

"Analytics Dashboard"

Describe it as:

"Historical Narrative Intelligence System"

Primary Demo:

Select a major event.

Show:

- Narrative evolution
- Sentiment evolution
- Entity growth
- Trend forecasting

The demo should feel like replaying history through data.

23. Success Criteria

Project is successful if:

- Complete ETL pipeline exists
- Star schema warehouse implemented
- Narrative tracking functional
- Sentiment analysis functional
- Predictive analytics functional
- Dashboard operational
- Historical intelligence demonstrated

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